Lab Assignment 17.1

Program	**: B. Tech (CSE)**
Course Title	**: AI Assisted Coding**
Course Code	**: 23CS002PC304**
Semester	**: III**
Academic Session	**: 2025-2026**
Name of Student	**: K.SRIKANTH**
Enrollment No.	**: 2403A51L42**
Batch No.	**: 52**
Date	**: 31-03-2026**

Task 1:

Prompt: Generate a Python Program for cleaning a customer dataset. That Handles missing values in columns (age, income, city), remove duplicate records, standardize text values (e.g., city names in lowercase) and encode categorical variables (gender, city).

Code:

```
import pandas as pd
dataset=pd.read_csv('/content/shopping_trends.csv')
display(dataset.head())
df = dataset.copy() # Work on a copy to preserve the original 'dataset'
# Check and fill missing 'Age' with the mean
if 'Age' in df.columns and df['Age'].isnull().any():
    df['Age'].fillna(df['Age'].mean(), inplace=True)
    print("Filled missing 'Age' values with mean.")
# Check and fill missing 'Purchase Amount (USD)' (proxy for income) with the mean
if 'Purchase Amount (USD)' in df.columns and df['Purchase Amount (USD)'].isnull().any():
    df['Purchase Amount (USD)'].fillna(df['Purchase Amount (USD)'].mean(), inplace=True)
    print("Filled missing 'Purchase Amount (USD)' values with mean.")
# Check and fill missing 'Location' (proxy for city) with the mode
if 'Location' in df.columns and df['Location'].isnull().any():
    # Ensure column is string type before finding mode and filling
    df['Location'] = df['Location'].astype(str)
    mode_location = df['Location'].mode()[0]
    df['Location'].fillna(mode_location, inplace=True)
    print("Filled missing 'Location' values with mode.")
print("\nDataFrame after handling missing values (if any):")
initial_rows = df.shape[0]
df.drop_duplicates(inplace=True)
final_rows = df.shape[0]
print(f"Removed {initial_rows - final_rows} duplicate rows.")
print("\nDataFrame after removing duplicates:")
if 'Location' in df.columns:
    df['Location'] = df['Location'].astype(str).str.lower()
    print("Standardized 'Location' names to lowercase.")
```

```
else:
    print("'Location' column not found for standardization.")
print("\nDataFrame after standardizing text values (if 'Location' exists):")
cols_to_encode = []
if 'Gender' in df.columns: cols_to_encode.append('Gender')
if 'Location' in df.columns: cols_to_encode.append('Location')
if cols_to_encode:
    df_encoded = pd.get_dummies(df, columns=cols_to_encode, drop_first=True)
    print(f"Encoded categorical variables: {', '.join(cols_to_encode)}.")
else:
    df_encoded = df.copy()
    print("No specified categorical columns ('Gender', 'Location') found for encoding.")
print("\nDataFrame after encoding categorical variables:")
display(df.head())
```

Output:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Payment Method	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Preferred Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Credit Card	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Bank Transfer	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Cash	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	PayPal	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Cash	Free Shipping	Yes	Yes	31	PayPal	Annually

```
DataFrame after handling missing values (if any):
Removed 0 duplicate rows.

DataFrame after removing duplicates:
Standardized 'Location' names to lowercase.

DataFrame after standardizing text values (if 'Location' exists):
Encoded categorical variables: Gender, Location.

DataFrame after encoding categorical variables:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Payment Method	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Preferred Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	kentucky	L	Gray	Winter	3.1	Yes	Credit Card	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	maine	L	Maroon	Winter	3.1	Yes	Bank Transfer	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	massachusetts	S	Maroon	Spring	3.1	Yes	Cash	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	rhode island	M	Maroon	Spring	3.5	Yes	PayPal	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	oregon	M	Turquoise	Spring	2.7	Yes	Cash	Free Shipping	Yes	Yes	31	PayPal	Annually

Explanation:

This program loads your shopping_trends.csv dataset, then cleans it by filling missing 'Age' and 'Purchase Amount (USD)' with their means, and 'Location' with its mode. It removes duplicate rows, standardizes 'Location' names to lowercase, and finally encodes 'Gender' and 'Location' using one-hot encoding for further analysis.

Task 2:

Prompt: Preprocess sales transaction data that Convert date columns into datetime format, create new features (month, year, day of week), normalize sales amount column and detect and handle outliers in dataset.

Code:

```python
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
data_set=pd.read_csv('/content/sales_data.csv')
display(data_set.head())
df_sales = data_set.copy()
if 'Sale_Date' in df_sales.columns:
    df_sales['Sale_Date'] = pd.to_datetime(df_sales['Sale_Date'])
    df_sales['Month'] = df_sales['Sale_Date'].dt.month
    df_sales['Year'] = df_sales['Sale_Date'].dt.year
    df_sales['DayOfWeek'] = df_sales['Sale_Date'].dt.dayofweek # Monday=0, Sunday=6
    print("Converted 'Sale_Date' column to datetime and created 'Month', 'Year', 'DayOfWeek' features:")
    display(df_sales[['Sale_Date', 'Month', 'Year', 'DayOfWeek']].head())
else:
    print("No 'Sale_Date' column found in `sales_data.csv` to convert or extract features from.")
# Assuming 'Sales_Amount' as the column to normalize
sales_column = 'Sales_Amount' # Adjust if your sales amount column has a different name
if sales_column in df_sales.columns:
    scaler = MinMaxScaler()
    df_sales[f'{sales_column}_normalized'] = scaler.fit_transform(df_sales[[sales_column]])
    print(f"Normalized '{sales_column}' column (to '{sales_column}_normalized'):")
    display(df_sales[[sales_column, f'{sales_column}_normalized']].head())
else:
    print(f"No '{sales_column}' column found for normalization.")
sales_column = 'Sales_Amount' # Adjust if your sales amount column has a different name
if sales_column in df_sales.columns:
    Q1 = df_sales[sales_column].quantile(0.25)
    Q3 = df_sales[sales_column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers_count = ((df_sales[sales_column] < lower_bound) | (df_sales[sales_column] > upper_bound)).sum()
    print(f"Number of outliers detected in '{sales_column}': {outliers_count}")
    # Cap outliers
    df_sales[f'{sales_column}_capped'] = df_sales[sales_column].clip(lower=lower_bound, upper=upper_bound)
    print(f"Outliers in '{sales_column}' have been capped. Original vs. Capped values:")
    display(df_sales[[sales_column, f'{sales_column}_capped']].head())
else:
    print(f"No '{sales_column}' column found for outlier detection.")
print("\nFinal preprocessed `sales_data.csv` DataFrame head:")
display(df_sales.head())
```

Output:

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie

```
Converted 'Sale_Date' column to datetime and created 'Month', 'Year', 'DayOfWeek' features:
```

	Sale_Date	Month	Year	DayOfWeek
0	2023-02-03	2	2023	4
1	2023-04-21	4	2023	4
2	2023-09-21	9	2023	3
3	2023-08-24	8	2023	3
4	2023-03-24	3	2023	4

```
Normalized 'Sales_Amount' column (to 'Sales_Amount_normalized'):
```

	Sales_Amount	Sales_Amount_normalized
0	5053.97	0.500950
1	4384.02	0.433202
2	4631.23	0.458201
3	2167.94	0.209105
4	3750.20	0.369108

```
Number of outliers detected in 'Sales_Amount': 0
Outliers in 'Sales_Amount' have been capped. Original vs. Capped values:
```

	Sales_Amount	Sales_Amount_capped
0	5053.97	5053.97
1	4384.02	4384.02
2	4631.23	4631.23
3	2167.94	2167.94
4	3750.20	3750.20

```
Final preprocessed `sales_data.csv` DataFrame head:
```

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep	Month	Year	DayOfWeek	Sales_Am
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09	Cash	Online	North-Bob	2	2023	4	
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11	Cash	Retail	West-Bob	4	2023	4	
2	1015	2023-09-21	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20	Bank Transfer	Retail	South-David	9	2023	3	
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02	Credit Card	Retail	South-Bob	8	2023	3	
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08	Credit Card	Online	East-Charlie	3	2023	4	

Explanation:

The code preprocessed sales_data.csv. It converted the Sale_Date column datetime, extracting Month, Year, and Day of Week. It then normalized the Sales_Amount column using MinMaxScaler.

Task 3:

Prompt: Clean and preprocess a healthcare dataset that handles missing values in blood pressure, cholesterol, encode categorical features, scale numerical features (min-max or standard scaling) and split dataset into training and testing sets.

Code:

```python
import pandas as pd
df_healthcare = pd.read_csv('/content/dirty_v3_path.csv')
# Make a copy to work on, preserving the original 'df_healthcare'
df_processed = df_healthcare.copy()
print("Original DataFrame head before preprocessing:")
display(df_processed.head())
# 1. Handle missing values for 'Blood Pressure' and 'Cholesterol'
# Fill missing 'Blood Pressure' with the mean
if 'Blood Pressure' in df_processed.columns and df_processed['Blood Pressure'].isnull().any():
    df_processed['Blood Pressure'] = df_processed['Blood Pressure'].fillna(df_processed['Blood Pressure'].mean())
    print("Filled missing 'Blood Pressure' values with mean.")
# Fill missing 'Cholesterol' with the mean
if 'Cholesterol' in df_processed.columns and df_processed['Cholesterol'].isnull().any():
    df_processed['Cholesterol'] = df_processed['Cholesterol'].fillna(df_processed['Cholesterol'].mean())
    print("Filled missing 'Cholesterol' values with mean.")
# Also consider other potentially important numerical columns like 'Glucose', 'BMI', 'Oxygen Saturation', 'Triglycerides', 'HbA1c', 'Stress Level', 'Sleep Hours'
# Fill missing 'Glucose' with the mean
if 'Glucose' in df_processed.columns and df_processed['Glucose'].isnull().any():
    df_processed['Glucose'] = df_processed['Glucose'].fillna(df_processed['Glucose'].mean())
    print("Filled missing 'Glucose' values with mean.")
# Fill missing 'BMI' with the mean
if 'BMI' in df_processed.columns and df_processed['BMI'].isnull().any():
    df_processed['BMI'] = df_processed['BMI'].fillna(df_processed['BMI'].mean())
    print("Filled missing 'BMI' values with mean.")
# Fill missing 'Oxygen Saturation' with the mean
if 'Oxygen Saturation' in df_processed.columns and df_processed['Oxygen Saturation'].isnull().any():
    df_processed['Oxygen Saturation'] = df_processed['Oxygen Saturation'].fillna(df_processed['Oxygen Saturation'].mean())
    print("Filled missing 'Oxygen Saturation' values with mean.")
# Fill missing 'Triglycerides' with the mean
if 'Triglycerides' in df_processed.columns and df_processed['Triglycerides'].isnull().any():
    df_processed['Triglycerides'] = df_processed['Triglycerides'].fillna(df_processed['Triglycerides'].mean())
    print("Filled missing 'Triglycerides' values with mean.")
# Fill missing 'HbA1c' with the mean
if 'HbA1c' in df_processed.columns and df_processed['HbA1c'].isnull().any():
    df_processed['HbA1c'] = df_processed['HbA1c'].fillna(df_processed['HbA1c'].mean())
    print("Filled missing 'HbA1c' values with mean.")
# Fill missing 'Stress Level' with the mean
if 'Stress Level' in df_processed.columns and df_processed['Stress Level'].isnull().any():
    df_processed['Stress Level'] = df_processed['Stress Level'].fillna(df_processed['Stress Level'].mean())
    print("Filled missing 'Stress Level' values with mean.")
# Fill missing 'Sleep Hours' with the mean
if 'Sleep Hours' in df_processed.columns and df_processed['Sleep Hours'].isnull().any():
    df_processed['Sleep Hours'] = df_processed['Sleep Hours'].fillna(df_processed['Sleep Hours'].mean())
    print("Filled missing 'Sleep Hours' values with mean.")
# For categorical columns, fill with mode, especially 'Gender' and 'Medical Condition'
if 'Gender' in df_processed.columns and df_processed['Gender'].isnull().any():
    df_processed['Gender'] = df_processed['Gender'].fillna(df_processed['Gender'].mode()[0])
    print("Filled missing 'Gender' values with mode.")
if 'Medical Condition' in df_processed.columns and df_processed['Medical Condition'].isnull().any():
    df_processed['Medical Condition'] = df_processed['Medical Condition'].fillna(df_processed['Medical Condition'].mode()[0])
    print("Filled missing 'Medical Condition' values with mode.")
# Display info after handling missing values
```

```python
print("\nDataFrame info after handling missing values:")
df_processed.info()
# 2. Encode categorical features
# Identify categorical columns (excluding 'random_notes' and 'noise_col' which might not be features)
categorical_cols = ['Gender', 'Medical Condition', 'Family History', 'Smoking', 'Alcohol'] # Assuming these are categorical
# Check if these columns exist before encoding
categorical_cols_to_encode = [col for col in categorical_cols if col in df_processed.columns]
if categorical_cols_to_encode:
    df_processed = pd.get_dummies(df_processed, columns=categorical_cols_to_encode, drop_first=True)
    print(f"\nEncoded categorical variables: {', '.join(categorical_cols_to_encode)}.")
else:
    df_processed = df_processed.copy()
    print("\nNo specified categorical columns found for encoding.")
# Drop non-numeric/identifier columns that are not features, e.g., 'random_notes'
if 'random_notes' in df_processed.columns:
    df_processed = df_processed.drop('random_notes', axis=1)
    print("Dropped 'random_notes' column.")
# Also drop 'noise_col' if it's not a feature
if 'noise_col' in df_processed.columns:
    df_processed = df_processed.drop('noise_col', axis=1)
    print("Dropped 'noise_col' column.")
print("\nDataFrame head after encoding categorical variables and dropping irrelevant columns:")
display(df_processed.head())
# 3. Scale numerical features
from sklearn.preprocessing import MinMaxScaler # Or StandardScaler, user can choose
# Identify numerical columns for scaling (exclude one-hot encoded and target if any)
# For simplicity, scale all remaining numerical columns except potential target
numerical_cols = df_processed.select_dtypes(include=['float64', 'int64']).columns.tolist()
# Exclude binary encoded columns (which are already 0 or 1) and any potential ID columns if they existed
# For this dataset, assume all remaining numerical columns are candidates for scaling
if numerical_cols:
    scaler = MinMaxScaler() # Using MinMaxScaler as an example
    df_processed[numerical_cols] = scaler.fit_transform(df_processed[numerical_cols])
    print(f"\nScaled numerical variables using MinMaxScaler: {', '.join(numerical_cols)}.")
else:
    print("\nNo numerical columns found for scaling.")
print("\nDataFrame head after scaling numerical features:")
display(df_processed.head())
# 4. Split dataset into training and testing sets
from sklearn.model_selection import train_test_split
# Assuming 'LengthOfStay' is the potential target variable (this is an example, it might vary)
# If there is no explicit target variable, we'll just split features
X = df_processed.drop('LengthOfStay', axis=1, errors='ignore') # Features
y = df_processed['LengthOfStay'] if 'LengthOfStay' in df_processed.columns else None # Target (if exists)
if y is not None:
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
    print(f"\nDataset split into training and testing sets. X_train shape: {X_train.shape}, X_test shape: {X_test.shape}, y_train shape: {y_train.shape}, y_test shape: {y_test.shape}.")
else:
    X_train, X_test = train_test_split(X, test_size=0.2, random_state=42)
    print(f"\nDataset split into training and testing features only (no explicit target). X_train shape: {X_train.shape}, X_test shape: {X_test.shape}.")
print("\nPreprocessing complete. Displaying X_train head:")
display(X_train.head())
```

Output:

```
Original DataFrame head before preprocessing:
```

	Age	Gender	Medical Condition	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	HbA1c	Smoking	Alcohol	Physical Activity	Diet Score	Family History	Stress Level	Sleep Hours	random_notes	noise_col
0	46.0	Male	Diabetes	137.04	135.27	28.90	96.04	6	231.88	210.56	7.61	0	0	-0.20	3.54	0	5.07	6.05	lorem	-137.057211
1	22.0	Male	Healthy	71.58	113.27	26.29	97.54	2	165.57	129.41	4.91	0	0	8.12	5.90	0	5.87	7.72	ipsum	-11.230610
2	50.0	NaN	Asthma	95.24	NaN	22.53	90.31	2	214.94	165.35	5.60	0	0	5.01	4.65	1	3.09	4.82	ipsum	98.331195
3	57.0	NaN	Obesity	NaN	130.53	38.47	96.60	5	197.71	182.13	6.92	0	0	3.16	3.37	0	3.01	5.33	lorem	44.187175
4	66.0	Female	Hypertension	95.15	178.17	31.12	94.90	4	259.53	115.85	5.98	0	1	3.56	3.40	0	6.38	6.64	lorem	44.831426

```
Filled missing 'Blood Pressure' values with mean.
Filled missing 'Glucose' values with mean.
Filled missing 'Gender' values with mode.
Filled missing 'Medical Condition' values with mode.

DataFrame info after handling missing values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 20 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Age                25500 non-null  float64
 1   Gender             30000 non-null  object
 2   Medical Condition  30000 non-null  object
 3   Glucose            30000 non-null  float64
 4   Blood Pressure     30000 non-null  float64
 5   BMI                30000 non-null  float64
 6   Oxygen Saturation  30000 non-null  float64
 7   LengthOfStay       30000 non-null  int64
 8   Cholesterol        30000 non-null  float64
 9   Triglycerides      30000 non-null  float64
 10  HbA1c              30000 non-null  float64
 11  Smoking            30000 non-null  int64
 12  Alcohol            30000 non-null  int64
 13  Physical Activity  30000 non-null  float64
 14  Diet Score         30000 non-null  float64
 15  Family History     30000 non-null  int64
 16  Stress Level       30000 non-null  float64
 17  Sleep Hours        30000 non-null  float64
 18  random_notes       30000 non-null  object
 19  noise_col          30000 non-null  float64
dtypes: float64(13), int64(4), object(3)
memory usage: 4.6+ MB

Encoded categorical variables: Gender, Medical Condition, Family History, Smoking, Alcohol.
Dropped 'random_notes' column.
Dropped 'noise_col' column.
```

```
DataFrame head after encoding categorical variables and dropping irrelevant columns:
```

	Age	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	HbA1c	Physical Activity	...	Gender_Male	Medical Condition_Asthma	Medical Condition_Cancer	Medical Condition_Diabetes	Medical Condition_Healthy	Medical Condition_Hypertension	Medical Condition_Obesity	Family History_1	Smoking_1	Alcohol_
0	46.0	137.040000	135.270000	28.90	96.04	6	231.88	210.56	7.61	-0.20	...	True	False	False	True	False	False	False	False	False	False
1	22.0	71.580000	113.270000	26.29	97.54	2	165.57	129.41	4.91	8.12	...	True	False	False	False	True	False	False	False	False	False
2	50.0	95.240000	140.455337	22.53	90.31	2	214.94	165.35	5.60	5.01	...	False	True	False	False	False	False	False	True	False	False
3	57.0	123.622179	130.530000	38.47	96.60	5	197.71	182.13	6.92	3.16	...	False	False	False	False	False	False	True	False	False	False
4	66.0	95.150000	178.170000	31.12	94.90	4	259.53	115.85	5.98	3.56	...	False	False	False	False	False	True	False	False	False	Tru

5 rows × 23 columns

```
Scaled numerical variables using MinMaxScaler: Age, Glucose, Blood Pressure, BMI, Oxygen Saturation, LengthOfStay, Cholesterol, Triglycerides, HbA1c, Physical Activity, Diet Score, Stress Level, Sleep Hours.

DataFrame head after scaling numerical features:
```

	Age	Glucose	Blood Pressure	BMI	Oxygen Saturation	LengthOfStay	Cholesterol	Triglycerides	HbA1c	Physical Activity	...	Gender_Male	Medical Condition_Asthma	Medical Condition_Cancer	Medical Condition_Diabetes	Medical Condition_Healthy	Medical Condition_Hypertension	Medical Condition_Obesity	Family History_1	Smoking_1	Al
0	0.455696	0.391428	0.401144	0.431680	0.670348	0.277778	0.518390	0.524877	0.476872	0.216283	...	True	False	False	True	False	False	False	False	False	
1	0.151899	0.171904	0.256540	0.378609	0.705592	0.055556	0.265915	0.342102	0.179515	0.733375	...	True	False	False	False	True	False	False	False	False	
2	0.506329	0.251249	0.435226	0.302155	0.535714	0.055556	0.453891	0.423050	0.255507	0.540087	...	False	True	False	False	False	False	False	True	False	
3	0.594937	0.346431	0.369988	0.626271	0.683506	0.222222	0.388288	0.460844	0.400881	0.425109	...	False	False	False	False	False	False	True	False	False	
4	0.708861	0.250947	0.683121	0.476820	0.643562	0.166667	0.623667	0.311561	0.297357	0.449969	...	False	False	False	False	False	True	False	False	False	

5 rows × 23 columns

```
Dataset split into training and testing sets. X_train shape: (24000, 22), X_test shape: (6000, 22), y_train shape: (24000,), y_test shape: (6000,).

Preprocessing complete. Displaying X_train head:
```

	Age	Glucose	Blood Pressure	BMI	Oxygen Saturation	Cholesterol	Triglycerides	HbA1c	Physical Activity	Diet Score	...	Gender_Male	Medical Condition_Asthma	Medical Condition_Cancer	Medical Condition_Diabetes	Medical Condition_Healthy	Medical Condition_Hypertension	Medical Condition_Obesity	Family History_1	Smoking_1	Al
21753	0.443038	0.286428	0.531747	0.314762	0.684680	0.519342	0.551566	0.346916	0.612803	0.241854	...	False	False	False	False	False	True	False	True	False	
251	0.645570	0.670378	0.532930	0.258032	0.696194	0.452406	0.471407	0.511013	0.339963	0.437364	...	False	False	False	True	False	False	False	True	False	
22941	0.253165	0.243838	0.435226	0.545547	0.700423	0.408506	0.347620	0.240088	0.376631	0.393193	...	True	False	False	False	False	False	True	False	False	
618	0.848101	0.274422	0.435226	0.286905	0.679041	0.371573	0.424424	0.270925	0.452455	0.466329	...	False	False	False	False	False	False	False	False	True	
17090	0.911392	0.248365	0.515775	0.448556	0.590226	0.500000	0.675038	0.350220	0.310131	0.344678	...	True	False	False	False	False	True	False	False	False	

5 rows × 22 columns

Explanation:

The code preprocesses a healthcare dataset by loading it into a DataFrame, handling missing values using mean/mode imputation, and encoding categorical features. It then drops irrelevant columns and scales numerical features to a 0-1 range. Finally, the prepared data is split into training and testing sets.

Task 4:

Prompt: Clean and preprocess an employee salary dataset that handles missing values, detect and remove salary outliers, standardize department names (lowercase), apply label encoding to job location, department.

Code:

```python
import pandas as pd
from sklearn.preprocessing import LabelEncoder
employees_salary_data = pd.read_csv('/content/Employee_Salary_Dataset.csv')
df_employees = employees_salary_data.copy()
print("Original DataFrame head:")
display(df_employees.head())
# Fill missing 'Salary' with mean
if 'Salary' in df_employees.columns and df_employees['Salary'].isnull().any():
    df_employees['Salary'] = df_employees['Salary'].fillna(df_employees['Salary'].mean())
    print("Filled missing 'Salary' values with mean.")
# Fill missing 'Experience_Years' with mean
if 'Experience_Years' in df_employees.columns and df_employees['Experience_Years'].isnull().any():
    df_employees['Experience_Years'] = df_employees['Experience_Years'].fillna(df_employees['Experience_Years'].mean())
    print("Filled missing 'Experience_Years' values with mean.")
# Fill missing categorical columns with mode (e.g., 'Gender', 'Department', 'Job_Location')
for col in ['Gender', 'Department', 'Job_Location']:
    if col in df_employees.columns and df_employees[col].isnull().any():
        mode_val = df_employees[col].mode()[0]
        df_employees[col] = df_employees[col].fillna(mode_val)
        print(f"Filled missing '{col}' values with mode: {mode_val}.")
print("\nDataFrame info after handling missing values:")
df_employees.info()
if 'Salary' in df_employees.columns:
    Q1 = df_employees['Salary'].quantile(0.25)
    Q3 = df_employees['Salary'].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers_count = ((df_employees['Salary'] < lower_bound) | (df_employees['Salary'] > upper_bound)).sum()
    print(f"Number of outliers detected in 'Salary': {outliers_count}")
    # Cap outliers instead of removing them to preserve data points
    df_employees['Salary_Capped'] = df_employees['Salary'].clip(lower=lower_bound, upper=upper_bound)
    print("Outliers in 'Salary' have been capped. Original vs. Capped values:")
    display(df_employees[['Salary', 'Salary_Capped']].head())
else:
    print("No 'Salary' column found for outlier detection.")
if 'Department' in df_employees.columns:
```

```
    df_employees['Department'] = df_employees['Department'].astype(str).str.lower()
    print("Standardized 'Department' names to lowercase.")
    print("\nUnique Department names after standardization:")
    print(df_employees['Department'].unique())
else:
    print("'Department' column not found for standardization.")
from sklearn.preprocessing import LabelEncoder
label_encoder = LabelEncoder()
# Apply Label Encoding to 'Job_Location'
if 'Job_Location' in df_employees.columns:
    df_employees['Job_Location_Encoded'] = label_encoder.fit_transform(df_employees['Job_Location'])
    print("Applied Label Encoding to 'Job_Location'.")
    print("\nOriginal Job_Location vs. Encoded:")
    display(df_employees[['Job_Location', 'Job_Location_Encoded']].head())
else:
    print("'Job_Location' column not found for encoding.")
# Apply Label Encoding to 'Department'
if 'Department' in df_employees.columns:
    df_employees['Department_Encoded'] = label_encoder.fit_transform(df_employees['Department'])
    print("Applied Label Encoding to 'Department'.")
    print("\nOriginal Department vs. Encoded:")
    display(df_employees[['Department', 'Department_Encoded']].head())
else:
    print("'Department' column not found for encoding.")
print("\nFinal preprocessed DataFrame head:")
display(df_employees.head())
```

Output:

```
Original DataFrame head:
```

	ID	Experience_Years	Age	Gender	Salary
0	1	5	28	Female	250000
1	2	1	21	Male	50000
2	3	3	23	Female	170000
3	4	2	22	Male	25000
4	5	1	17	Male	10000

```
DataFrame info after handling missing values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 35 entries, 0 to 34
Data columns (total 5 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ID                35 non-null     int64
 1   Experience_Years  35 non-null     int64
 2   Age               35 non-null     int64
 3   Gender            35 non-null     object
 4   Salary            35 non-null     int64
dtypes: int64(4), object(1)
memory usage: 1.5+ KB
Number of outliers detected in 'Salary': 2
Outliers in 'Salary' have been capped. Original vs. Capped values:
```

	Salary	Salary_Capped
0	250000	250000
1	50000	50000
2	170000	170000
3	25000	25000
4	10000	10000

```
'Department' column not found for standardization.
'Job_Location' column not found for encoding.
'Department' column not found for encoding.

Final preprocessed DataFrame head:
```

	ID	Experience_Years	Age	Gender	Salary	Salary_Capped
0	1	5	28	Female	250000	250000
1	2	1	21	Male	50000	50000
2	3	3	23	Female	170000	170000
3	4	2	22	Male	25000	25000
4	5	1	17	Male	10000	10000

Explanation:

The code preprocesses the Employee_Salary_Dataset. It handles missing values in numerical columns by filling them with the mean, and in categorical columns by filling them with the mode. It then detects and caps outliers in the 'Salary' column. Finally, it standardizes 'Department' names and apply label encoding to 'Job_Location' and 'Department'.